--2-D ARRAYS USING FUNCTIONS--

1. ***WAP to find sum of odd and even elements in m*n matrix***

```
#include<stdio.h>
#include<conio.h>

void input( int [100 ][100] , int, int);
void display(int [ 100][100], int, int);
void sum_odd_even(int [ 100][100], int, int);

int main(){

        int a[100][100], row, column;
        printf("Enter the row of matrix:");
        scanf("%d", &row);
        printf("Enter the column of matrix:");
        scanf("%d", &column);
        input(a, row, column);
        printf("\nThe Array Elements are:\n");
        display(a, row, column);
        sum_odd_even( a, row, column);
   getch();
   return 0;
}


void input( int a[ 100][100] , int m, int n){

        int i, j;
        for(i=0;i<m; i++){
                for(j=0;j<n; j++){
                        printf("Enter a[%d][%d] element:",i, j);
                        scanf("%d", &a[i][j]);

                }
        }
}
```

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```
void display( int a[ 100][100] , int m, int n){
        int i, j;
        for(i=0;i<m; i++){
                for(j=0;j<n; j++){
                        printf("%d\t", a[i][j]);
                }
                printf("\n");
        }
}

void sum_odd_even(int a[ 100][100], int m, int n){
                int i, j, osum=0,esum=0;
        for(i=0;i<m; i++){
                for(j=0;j<n; j++){
                        if(a[i][j]%2==0){
                                esum=esum+a[i][j];
                        }
                        else{
                                osum=osum+a[i][j];
                        }
                }
        }
        printf("\nEven Sum=%d and Odd Sum=%d", esum, osum);
}
```

2. **WAP to find sum of main diagonal and sum of right diagonal of n*n square matrix.**

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```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>

void input( int [100 ][100] , int);
void display(int [ 100][100], int);
void sum_diagonal(int [ 100][100], int);
int main(){
        int a[100][100], row, column;
        printf("Enter the row of matrix:");
```

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```
        scanf("%d", &row);
        printf("Enter the column of matrix:");
        scanf("%d", &column);

        //for square matrix rows should be equal to columns
        if(row!=column){
                printf("Matrix should be square.");
                exit(0);
        }

        input(a, row);
        printf("\nThe Array Elements are:\n");
        display(a, row);
        sum_diagonal ( a, row);
    getch();
    return 0;
}

void input( int a[ 100][100] , int m){
        int i, j;
        for(i=0;i<m; i++){
                for(j=0;j<m; j++){
                printf("Enter a[%d][%d] element:",i, j);
                scanf("%d", &a[i][j]);

                }
        }
}

void display( int a[ 100][100] , int m){
        int i, j;
        for(i=0;i<m; i++){
                for(j=0;j<m; j++){
                        printf("%d\t", a[i][j]);
                }
                printf("\n");
        }
}
```

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```
void sum_diagonal(int a[ 100][100], int m){
        int i, j, main_diagonal=0, right_diagonal=0;

        for(i=0; i<m; i++){
                for(j=0; j<m; j++){
                        if(i==j){
                                main_diagonal = main_diagonal +a[i][j];
                        }
                        if((i+j)==(m-1)){
                                right_diagonal = right_diagonal +a[i][j];
                        }
                }
        }
        printf("\nThe sum of main diagonal elements is:%d", main_diagonal);
        printf("\nThe sum of right diagonal elements is:%d", right_diagonal);
}
```

3. WAP to find highest and lowest element of m*n matrix..

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>

void input( int [100 ][100] , int, int);
void display(int [ 100][100], int, int);
void max_min(int [ 100][100], int, int );
```

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```
int main(){
        int a[100][100], row, column, max, min;
        printf("Enter the row of matrix:");
        scanf("%d", &row);
        printf("Enter the column of matrix:");
        scanf("%d", &column);
        input(a, row, column);
        printf("\nThe Array Elements are:\n");
        display(a, row, column);
        max_min( a, row, column);
   getch();
   return 0;
}
```

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```
void input( int a[ 100][100] , int m, int n){
        int i, j;
        for(i=0;i<m; i++){
                for(j=0;j<n; j++){
                        printf("Enter a[%d][%d] element:",i, j);
                        scanf("%d", &a[i][j]);

                }
        }
}


void display(int a[ 100][100] , int m, int n){
        int i, j;
        for(i=0;i<m; i++){
                for(j=0;j<n; j++){
                        printf("%d\t", a[i][j]);
                }
                printf("\n");
        }
}

void max_min(int a[ 100][100] , int m, int n){
        int i, j, max=a[0][0], min=a[0][0];

        for(i=0; i<m; i++){
                for(j=0; j<n; j++){

                        if(a[i][j]>max){
                                max=a[i][j];
                        }
                        if(a[i][j]<min){
                                min=a[i][j];;
                        }
                }
        }
        printf("\nMaximun is:%d and Minimum is:%d", max, min);
}
```

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4. *WAP to add two n*n square matrix*

```
#include<stdio.h>
#include<conio.h>

void input( int [100 ][100] , int);
void display(int [ 100][100], int);
void add(int[100][100], int[100][100],int[100][100] ,int);

int main(){
        int a[100][100], b[100][100],c[100][100], n;
        printf("Enter size  of square matrix:");
        scanf("%d", &n);

        printf("\nEnter First Matrix:\n");
        input(a, n);

        printf("\nEnter Second Matrix:\n");
        input(b, n);

        printf("\nFirst matrix:\n");
        display(a, n);

        printf("\nSecond matrix :\n");
        display(b, n);

        add(a, b, c,n);
        printf("\nAdded matrix:\n");
        display(c, n);
   getch();
   return 0;
}

void input( int a[ 100][100] , int n){
        int i, j;
        for(i=0;i<n; i++){
                for(j=0;j<n; j++){
                        printf("Enter a[%d][%d] element:",i, j);
                        scanf("%d", &a[ i ][ j ]);
                }
        }
}
```

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```
void display( int a[ 100][100] , int n){
        int i, j;
        for(i=0;i<n; i++){
                for(j=0;j<n; j++){
                        printf("%d\t", a[i][j]);
                }
                printf("\n");
        }
}

void add(int a[100][100], int b[100][100],int c[100][100] ,int n){

int i, j;
        for(i=0;i<n; i++){
                for(j=0;j<n; j++){
                        c[i][j]=a[i][j]+b[i][ j];
                }
        }
}
```

5. *WAP to multiply two m*n square matrix*

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>

void input( int [100 ][100] , int, int);
void display(int [ 100][100], int, int);
void mul(int[100][100], int[100][100],int[100][100] ,int, int);
int main(){
        int a[100][100], b[100][100],c[100][100], r1, c1, r2,c2;
        printf("Enter the row of first matrix:");
        scanf("%d", &r1);
        printf("Enter the column of first matrix:");
        scanf("%d", &c1);
        printf("Enter the row of second matrix:");
        scanf("%d", &r2);
        printf("Enter the column of second matrix:");
        scanf("%d", &c2);
        if(c1!=r2){
                printf(“Matrix multiplication not possible.”);
                exit(0);
        }
```

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```
        printf("\nEnter First Matrix:\n");
        input(a, r1, c1);

        printf("\nEnter Second Matrix:\n");
        input(b, r2, c2);

        printf("\nFirst array elements are:\n");
        display(a, r1, c1);

        printf("\nSecond array elements are:\n");
        display(b, r2, c2);

        mul(a, b, c, r1, c2);
        printf("\nMultiplied array elements are:\n");
        display(c, r1, c2);
    getch();
    return 0;
}


void input( int a[ 100][100] , int m, int n){
        int i, j;
        for(i=0;i<m; i++){
                for(j=0;j<n; j++){
                        printf("Enter a[%d][%d] element:",i, j);
                        scanf("%d", &a[i][j]);
                }
        }
}

void display( int a[ 100][100] ,int m,  int n){
        int i, j;
        for(i=0;i<m; i++){
                for(j=0;j<n; j++){
                        printf("%d\t", a[i][j]);
                }
                printf("\n");
        }
}
```

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```
void mul(int a[100][100], int b[100][100],int c[100][100] ,int m, int n){

int i, j,k, sum=0;
        for(i=0; i<m; i++){
                for(j=0; j<n; j++){
                        for(k=0; k<n; k++){
                                        sum=sum+a[i][k]*b[k][j];
                        }
                        c[i][j]=sum;
                        sum=0;
                }
        }
}
```

6. *WAP to find transpose of m*n square matrix*

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>

void input( int [100 ][100] , int, int);
void display(int [ 100][100], int, int);
void transpose(int[100][100] ,int[100][100], int, int);
int main(){
        int a[100][100], b[100][100],  m, n;
        printf("Enter the row of  matrix:");
        scanf("%d", &m);
        printf("Enter the column of matrix:");
        scanf("%d", &n);

        printf("\nEnter Matrix:\n");
        input(a, m, n);

        printf("\nThe matrix is:\n");
        display(a, m, n);

        transpose(a, b, m, n);
        printf("\nTranpose matrix is :\n");
        //for transpose matrix row=n and column=m
        display (b, n, m);
   getch();
   return 0;
}
```

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```
void input( int a[ 100][100] , int m, int n){

	int i, j;
	for(i=0;i<m; i++){
		for(j=0;j<n; j++){
			printf("Enter a[%d][%d] element:",i, j);
			scanf("%d", &a[i][j]);
		}
	}
}


void display( int a[ 100][100] ,int m,  int n){

	int i, j;
	for(i=0;i<m; i++){
		for(j=0;j<n; j++){
			printf("%d\t", a[i][j]);
		}
		printf("\n");
	}
}



void transpose(int a[100][100], int b[100][100], int m, int n){

	int i, j;
	for(i=0; i<m; i++){
		for(j=0; j<n; j++){
			b[j][i]=a[i][j];
		}
	}
}
```

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